

Effect of Certain established Ayurvedic Rasayana Food Supplements on Aging Process at Antarctica – A review

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INTRODUCTION

It is a well-known fact that stress generates free radicals that may cause lipid peroxidation and affect the antioxidant status of the body. The body has certain set mechanisms to exert scavenging action on free radicals that include some antioxidant enzyme systems like super oxide dismutase, catalase and glutathione peroxidase, glutathione-s-transferase and glutathione reductase which are involved in the detoxification of reactive oxygen species directly or indirectly. Low levels of these antioxidant enzymes will cause accumulation of free radicals and may finally actively participate in the process of tissue injury (Mulden et al, 1991). According to another study free radicals damage tissue both at cellular and sub-cellular level by causing lipid peroxidation in the cell membranes and the inactivation of membrane bound enzymes (Florence, 1995).

India has been arranging expeditions to Antarctica for the last two decades. An Antarctic expeditioner has to face and cope up with many physical and psychological stresses apart from the stress related to UV radiations, magnetic field, high wind velocity, extreme cold conditions, and circadian biorhythms. Men living in Antarctica also suffer significant emotional strain as a result of physical isolation and social deprivation in addition to physiological stress imposed by few factors mentioned earlier (Roy and Dev, 1999). At the same time exposure to extreme cold poses stress and causes a number of metabolic disturbances in addition to problems of frostbite. So all these stress factors cause oxidative stress, which can be induced either by depletion of antioxidants and / or by an overload of oxidant species so that antioxidant levels become insufficient (Halliwell and Gutteridge, 1990). In such a situation it is very important to supplement

antioxidants to keep up antioxidant status of the body. Attempts have been made to study the mechanisms that might be involved in man's physiological responses to polar conditions. Few of them have helped in evaluating the nutritional requirements of men working in Polar Regions.

Ayurveda the first recorded scientific medicine in the history of the world emphasizes on prevention of disease and promotion of health by strengthening the tissue so that they can withstand the exogenous and endogenous stressors. This is achieved by modulating diet and life style as well as by the appropriate use of the phytomedicine/drugs that restore the equilibrium of the body (Dahanukar and Thatte, 1989a, 1989b). Over 600 plants have been described in various Ayurvedic texts like *Charak Samhita*, *Sushruta Samhita* and *Nighantu*. There are interesting groups of herbs called *rasayana* herbs that have proven adaptogenic properties. The word *RASAYANA* means the path that plasma follows (*RASA*–Plasma; *AYANA*–Path). It is believed that in Ayurveda the qualities of *RASA DHATU* (*plasma tissue*) influence the health of other dhatu's (Tissues) of the body. Hence any medicine that improves the quality of *RASA* should strengthen or promote the health of all tissues of body. So these *rasayana* herbs are said to prevent aging, re-establish youth, strengthen life and brain power and prevent disease all of which imply that they increase the resistance of body against any onslaught and act as non-specific adaptogen (Ghaniker, 1981).

Although there is a provision of very good nutritional diet to the members of the expedition, yet the scientific field lacks the data related to nutritional requirements in stressful Antarctic climate. Keeping the above view in mind, a food supplement of various *rasayana* herbs was prepared and a study was carried out to look into the adaptogenic effects of *rasayana* herbs on aging process at Antarctica.

SUBJECTS AND METHODS

The study was conducted on Indian station "Maitri" on subjects out of the 23rd Indian Scientific Expedition to Antarctica. Out of total 50 team members, 21 members volunteered for supporting this study. All the members were briefed about the project and informed consent was taken to join the study as per international ethical guidelines. The trial drug was tested for heavy metals and it was free from cadmium, lead and arsenic. The food supplement was also tested for microbial load and total bacterial count was 7.3×10^4 CFU/g whereas total fungal count, enterobacteria count and salmonella sp. was within permissible limits. All the subjects were provided

with the food supplements *AYUSH POSHAK PEYA* (APP 125ml) and *AYUSH POSHAK YOGA* (APY 50g) once daily. The subjects who could not take the drug regularly were put under dropouts. Out of 21 subjects, 8 were dropped out due to different reasons. In addition to it a control group consisting of 7 subjects was maintained throughout the study period which was not fed with the supplement. The sampling for all the parameters was done at zero day and fortnightly intervals up to 6 weeks. Biochemical parameters were measured using standard methods to assess the antioxidant, antistress and adaptogenic effect of food supplement.

RESULTS

A significant variation (t stat. $> t$ critic at $p < .05$) in levels of catalase was observed through out the study (Fig.1). The levels were highest at 4th week and it was lower in control group as compared to treatment group. There was slight decline in levels at 6th week. Starting from 0 day there was a gradual increase in catalase levels except a slight fall at 6th week in control group whereas in treatment group it abruptly shoot up at 4th week and declined at 6th week. The trend of increase in level was same in both the groups. Level of malondialdehyde, which is a marker of lipid peroxidation, was found to be significantly higher (t stat. $> t$ critic at $p < .05$) in control group subjects as compared to subjects under trial drug (Fig.2). The levels increased abruptly at 2nd week and declined in similar manner at 6th week. At zero day level malondialdehyde was 21 ± 4.37 nmole/ml (already higher as compared to normal population range) whereas the levels came down to nearly normal levels at 6th week in treatment group. Malondialdehyde levels were nearly equal at zero day as well as 6th week in control group subjects; however there was abrupt increase at 2nd week with slight increase at 4th week stage followed by abrupt fall to nearly zero day level at 6th week stage. The concentration of vitamin E was significantly less (t stat. $> t$ critic at $p < .05$) in control group as compared to treatment group (Fig.3). Initially concentration of vitamin E was 5.96 ± 0.88 mg/L in control group and it showed gradual increase in treatment group in 6th week stage up to a level of 10.06 ± 1.66 mg/L whereas the increase in concentration was very less in the control group. At 2nd week, rather it demonstrated a slight decrease followed by a regain at 4th and 6th week intervals. Levels of superoxide dismutase showed a gradual increase starting with 4.38 ± 0.50 U/L at 0 day to 5.14 ± 0.71 U/L at the end of study in treatment group, whereas the levels significantly decreased (t stat. $> t$ critic at $p < .05$) in the control group up to 2.63 ± 0.41 U/L at the 6th week (Fig. 4).

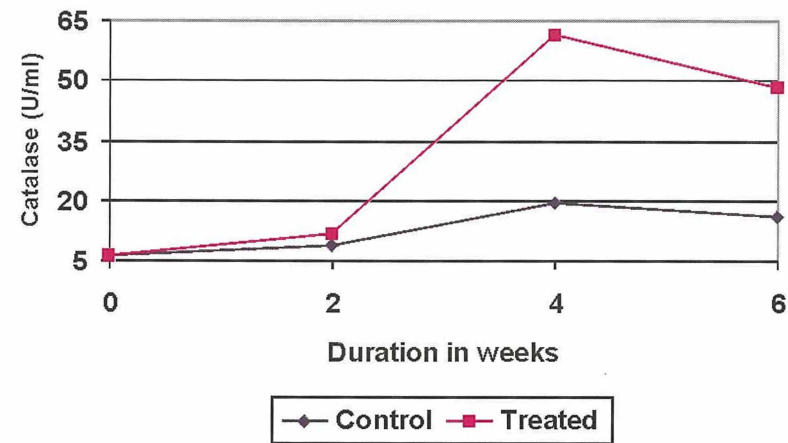


Fig. 1: Effect of APY and APP on Catalase

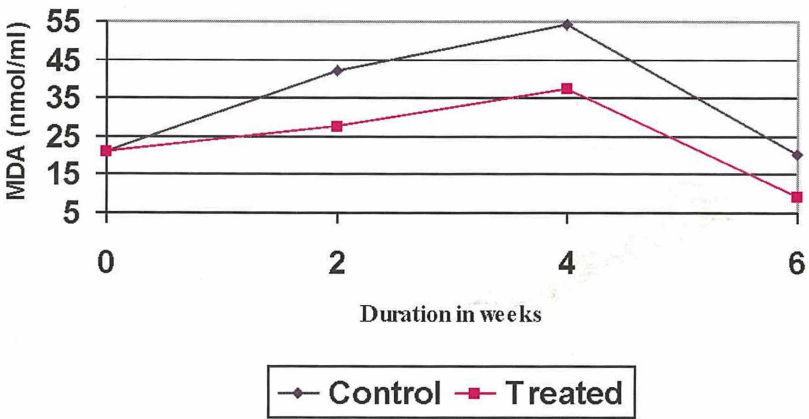


Fig. 2: Effect of APY and APP on Malondialde hyde (MDA)

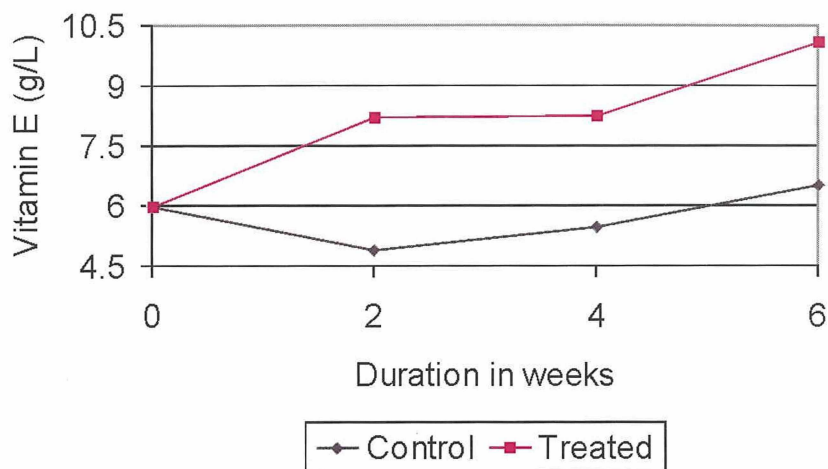


Fig. 3: Effect of APY and APP on Vitamin E

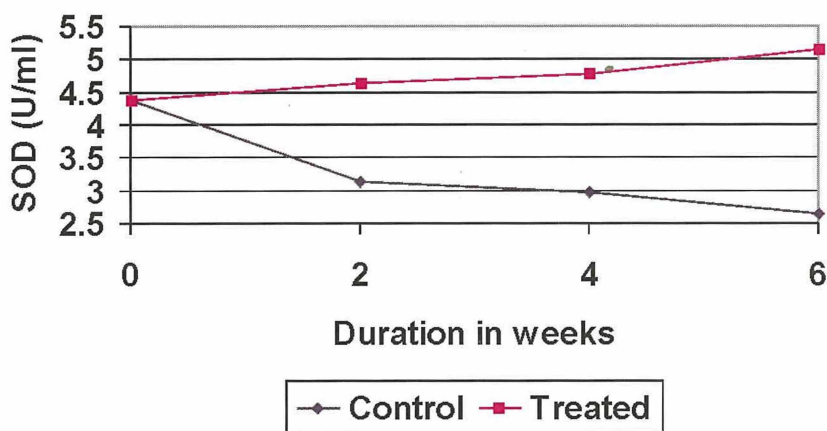


Fig. 4: Effect of APY and APP on Superoxidedismutase (SOD)

Concentration of immunoglobulin IgM was 80.75 ± 30.38 at zero day (Tab. 1). At 2nd week the level decreased in both the groups, however the concentration was almost 1/3rd in treatment group subjects as compared to the levels in control subjects. At 4th week stage the level increased abruptly in all subjects. The levels increased to 106 ± 8.94 at 4th week and 115 ± 9.35 at 6th week in control subjects (even higher than at 0 day) whereas the values

were 46.15 ± 11.39 and 55.38 ± 15.34 at respective stages in treatment group. On the whole the pattern of levels of IgM were similar in control as well as treatment group subjects at all the stages.

The concentration of immunoglobulin IgA was 146 ± 15.96 at 0 day (Table 1). The levels increased continuously at 2nd week and 4th week stage in control group. Level of IgA was similar at 4th and 6th week in control group. In contrast to it the levels decreased in treatment group subjects at 2nd week but depicted slight increase at 4th and 6th week as compared to levels at 2nd week stage. On the whole the levels of these immunoglobulins were lower at all stages as compared to concentration at 0 day in treatment group whereas the concentration was higher in control group.

The concentration of IgG was very high to the tune of 3091 ± 705 at 0 day (Table 1). The level was decreased at 2nd week, increased at 4th week and again decreased at 6th week stage in control group subjects. The level of IgG was lower in treatment group as compared to control subjects at all stages except at 6th week where it was higher in treatment group. The level of IgG decreased till 4th week and slightly increased at 6th week in treatment group.

Table 1: Effect of APY and APP on Immunoglobulin

Immunoglobulin	0 day	2nd week		4th week		6th week	
		Control	Treated	Control	Treated	Control	Treated
	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD
IgA (mg/dl)	146.1 $\pm$ 15.96	187.5 $\pm$ 31.58	121.67 $\pm$ 25.26	244.4 $\pm$ 22.51	131.15 $\pm$ 31.3	244 $\pm$ 23.02	123.08 $\pm$ 28.32
IgG (mg/dl)	3091.3 $\pm$ 705.76	1780 $\pm$ 132.57	1599.08 $\pm$ 117.33	2600 $\pm$ 106.07	1542.31 $\pm$ 252.36	1422 $\pm$ 263.95	1705.38 $\pm$ 189.76
IgM (mg/dl)	80.75 $\pm$ 30.39	48.75 $\pm$ 8.33	16.46 $\pm$ 2.98	106 $\pm$ 8.94	46.15 $\pm$ 11.39	115 $\pm$ 9.35	55.38 $\pm$ 15.34

The food supplement was of rich nutritional value so an attempt was made to ascertain the effect of food supplement on lipid profile (Table 2). Results show that the level of total cholesterol was lowered gradually from 0 day to 6th week in treatment group ($p > 0.05$). The pattern was almost similar in control group except slight increase at 2nd week and then an abrupt fall at 4th week followed by a slight increase at 6th week. The level of HDL was higher in the treatment group at all the stages as compared to control

group, however it demonstrated a slight increase at 2nd week ($p>0.05$) followed by slight decline at 4th and 6th week. The level of LDL cholesterol was 158 ± 19.94 at 0 day. In control group, it further increased to 164.8 ± 18.5 at 2nd week ($p>0.05$), decreased at 4th week stage followed by no significant increase at 6th week. In contrast to it, the level of LDL decreased continuously to a level of 126.77 ± 31.43 at 6th week stage in treatment group. Level of VLDL cholesterol was 21.10 ± 3.1 at 0 day and it continued to be nearly same till end except slight decline at 4th week in control group, however the values decreased as compared to 0 day values throughout the study except a slight increase at 2nd week.

Table 2: Effect of APY and APP on Lipid Profile

	0 day	2nd week		4th week		6th week	
	<u>Mean±SD</u>	Control	Treated	Control	Treated	Control	Treated
		<u>Mean±SD</u>	<u>Mean±SD</u>	<u>Mean±SD</u>	<u>Mean±SD</u>	<u>Mean±SD</u>	<u>Mean±SD</u>
Cholesterol (mg/dl)	232.22±19.9	249.17±10.3	222±28.73	188.5±17.31	210.15±36.3	208.5±30.71	202.23±30.98
HDL (mg/dl)	52±8.86	60.67±6.8	59.92±6.39	43.5±6.24	57.69±6.14	52.95±12.13	56.27±8.37
LDL (mg/dl)	158±19.94	164.83±18.57	134.58±22.28	139.25±33.6	132.15±34.7	131.75±18.25	126.77±31.43
HDL/LDL ratio	0.336±0.081	0.375±0.078	0.455±0.082	0.338±0.143	0.467±0.137	0.407±0.101	0.476±0.161
VLDL (mg/dl)	21.1±3.18	23.67±5.39	25.92±6.37	20.75±5.85	21.08±5.02	24.25±13.96	19.62±5.28
Triglyceride (mg/dl)	112.4±16.26	118.17±25.4	130.25±31.37	103.5±29.1	101.23±26.15	121.25±67.94	97.54±27.28

Level of triglycerides was 112.4 ± 16.26 at 0 day. It increased at 2nd week stage in both the groups; however the increase was more in treatment group as compared to control group. At 4th week stage level of triglyceride was lowered down in both groups and again a slight higher level was demonstrated in trial drug group. The level of triglycerides increased abruptly at 6th week in control group whereas the level was similar to 4th week in treatment group. HDL/LDL ratio was 0.33 at 0 day. It increased slightly (about 10%) in control group subjects at 2nd and 4th week with further increase of about 10% at 6th week. The ratio of HDL/LDL increased to the tune of 25% in treatment group at 2nd week and was maintained at same

increased level till 6th week. At all stages the ratio was higher in treatment group.

DISCUSSION

In present study vitamin E levels were found significantly lower in control subjects as compared to trial drug group subjects, which have been reflected by the levels of malondialdehyde also. Vitamin E may trap chain reaction started by oxygen radicals of endogenous or exogenous origin thus protecting cells from lipid peroxidation. Vitamin E is known to prevent the formation of peroxides and it was suggested that Vitamin E may prevent the peroxidation of arachidonate in platelets perhaps at the step of phospholipase or may inhibit PGI-2 synthetase. According to a recent study low serum concentrations have been correlated with higher degree of lipid peroxidation (Malvy et al, 1994). This has been reflected at second week stage in control group in which serum levels have gone down as compared to 0 day stage in present study. Higher levels of malondialdehyde than normal levels at zero day level seem to depict that all the subjects had already undergone stress. The levels were higher in control group subjects at all stages demonstrating more production of free radicals due to stress. The levels were significantly lower in trial drug group subjects probably due to the presence of antioxidants in the food supplement. An abrupt increase of malondialdehyde level at 2nd week stage may be due to higher level of stress in earlier days of arriving at Antarctica and non-acclimatization in Antarctic environment, however the fall in levels of malondialdehyde at 6th week stage depict the acclimatization of all the subjects till end of the study. The levels of malondialdehyde were just near to normal in trial drug subjects, which could be due to antioxidant effects of trial drug on the subjects. During a polar expedition an increase in production of malondialdehyde was measured in erythrocytes and plasma followed by decreased blood concentration of vitamin E (Satija et al, 2000). The levels of SOD decreased in subjects of control group throughout the study however the levels were significantly increased in subjects under trial drug. The results clearly depict that control group subjects are under oxidative stress however the raised levels in trial drug subjects demonstrate the tendency of development of scavenging mechanism by the body against the increased level of free radicals. The levels of catalase followed the trend followed by malondialdehyde levels. The lower level of catalase in control group subjects could be due to inactivation of the enzymes due to free radicals because Florence (1995) has demonstrated that free radicals damage tissue both at cellular and sub cellular level by

causing peroxidation in cell membranes and the inactivation of membrane bound enzymes.

The antioxidant effect of drug can be attributed to its components that are having established antioxidant properties. In different studies Piper longum has been shown to have free radical scavenging activity, antibacterial, antioxidant activity (Thirunarayan, 2001), to increase antioxidant enzyme activity of SOD, Catalase, and glutathione and decrease lipid peroxidation (Krishnamurthy, 2003). Almonds are rich source of natural vitamin E and are having antioxidant property due to efficient utilization by the body (Chopra, 2004). Ashwagandha have been shown to exert antistress adaptogenic activity (Singh, 1982), increase succinate dehydrogenase enzyme levels in brain tissue indicating facilitation of energy formation and conservation due to stress (Singh, 1999), increase superoxide dismutase levels and catalase levels significantly (Salil, 1997). The herbal drug guduchi has been reported to inhibit lipid peroxidation (Kanwaljeet and Singh, 1994; Goel, 2002), exert antioxidant property (Mathew and Kuttan, 1997, Deshmukh and Usha, 1994, Desai et al, 2002) and remarkable antistress activity in several studies (Sharma et al, 1996) Guduchi is known to be rich source of trace element zinc and copper that protect cells from the damaging effects of oxygen radicals generated during immune activation (Shankar and Parsad, 1998).

The lower levels of IgG in treatment group indicate better resistance developed in the body as an adaptive mechanism to stress. At 4th week stage the levels in control group were raised significantly indicating the effect of stress on body however in treatment group the lower level depict more tolerance of the body. The levels of secretory immunoglobulin IgA also depicts development of tolerance in treatment group subjects in stress conditions. Concentration of IgM was decreased at 2nd week in both the groups as compared to 0 day values which shows that the body became less prone to infections or it could be effect of bacteria free environment of Antarctica on immune system. The concentration of the immunoglobulin increased gradually till 6th week stage in both groups however the values were significantly lower in treatment group as compared to control group probably due to development of better resistance in treatment group against stress induced changes in immunoglobulin levels. Amongst many medical problems immuno-suppression has been observed during Antarctica sojourn (Tashpulatov, 1974, Muchmore et al, 1970, Muller et al, 1995a, 1995b). There is a wide body of literature demonstrating the detrimental effects of stress on the immune system (Tingate et al, 1997, Boneau et al, 1990, Khansari et

al, 1990). It has been recognized that stress activates corticotrophin releasing factor (CRF) leading to secretion of adrenocorticotrophin releasing hormone (ACTH) and the opioid peptides beta lipotropin/beta endorphin from the anterior pituitary. So these peptides activate immune events in the body (Morley, 1981). The changes in level of these immunoglobulins in this study are in accordance with earlier Antarctic studies in which Muchmore et al (1970) found a decline in serum IgG and IgM, whereas Tashpulativ (1974) observed a rise in serum IgG and IgM levels in Antarctic workers. The role of antioxidants in the immune system is multifaceted and well known as these can serve to either suppress or enhance the immune response. Depending on the desired response different antioxidants can play different roles in balancing the immune response effectively in an individual. Beta-carotene, selenium, vitamin C and vitamin E are endogenous antioxidants that work to eliminate free radicals from the body. In fact many immune cells such as phagocytes produce free radicals in order to destroy an antigen. If an individual suffers tissue damage and at the same time due to infection phagocytes are releasing free radicals to destroy an antigen, the Fenton reaction will produce hydroxyl radical because the metals released from damaged tissue would react with the H_2O_2 released from the phagocyte. In order to clear these oxidants there must be balance for optimal response. For too many antioxidants in our diet would limit the immune cells, while not enough would cause phagocytes to produce so many radicals, that they would kill themselves and again be ineffective against any antigen present. In present study there was stress on all subjects and free radicals were probably generated (evident from the elevated serum MDA levels) however due to food supplement in treatment group, the level of antioxidants was enough to give resistance to stress related changes in immunoglobulins. As *Amygdalus prunus* (Badam) is rich source of natural vitamin E and has been part of the food supplement, it would have supplied vitamin E in sufficient quantity to meet requirements of the body. Many of the adaptogen rasayana have already been shown to possess immuno-stimulant activity (Wagner et al, 1994). Whether this immuno-stimulation is responsible for the adaptogenic potential is not clear till date. As mentioned earlier during stress, prostaglandin and antioxidant systems of target organs serve as natural defence mechanism to counteract the deleterious effect of stress-induced lipid peroxidation in the target organ. In case of its depletion, stress manifestations occur, however prevention of their depletion may be a mechanism for counteracting stress and the beginning of the adaptation process (Meerson, 1984). *Tinospora*

cordifolia (Guduchi) has been shown to exert remarkable antioxidant property (Mathew and Kuttan, 1997), antistress activity, Immunostimulant (Deshmukh and Usha, 1994) immunomodulatory effects (Dubey et al, 2002). Ashwagandha has been shown to increase non-specific general immunity in children (Parbhakar et al., 1994). It contains withanolides (Withaferin A, Withanolides D, and F) that have been reported to possess both immunosuppression and immunostimulatory properties of activating immune responses in various test studies (Uniyal, 2002). Withania somnifera glycowithanolides (WSG) induced a dose related increase in SOD, CAT and glutathione peroxidase activities in frontal cortex and striatum of rats. The increase was statistically significant only after 14 or 21 days of treatment in SOD and Catalase. Effect of lower dose of WSG on glutathione peroxidase activity was only evident after 21 days treatment and findings indicate that increase in oxidative free radical scavenging activity of WSG may be responsible for the antistress immunomodulatory and antiaging effects (Salil, 1997). The similar types of trends have been found in the present study. Phenyl propene diasaccharides (Cordifolioside A and B) have been identified as active principles for immunostimulant action of the herb. It also decreases the incidence of septicaemia resulting in enhanced survival time. Pippali has also been hypothesized as a drug possibly acting through some cidal constituents by improving immune status of patient (Abbas et al, 1997).

The higher levels of cholesterol at zero and 2nd week stage may be due to emotional stress encountered by the members during journey period and at initial stages of adjustment in Antarctica, whereas the fall in levels at 4th and 6th week could be due to adjustment and acclimatization of the members at later stages. It has already been demonstrated that emotional stress is accompanied by an increase in serum cholesterol. Exposure to cold has also been shown to raise the serum cholesterol level (Satija et al, 2000). Further more an increased craving for dietary fat in extreme cold climate has been reported. It may therefore be expected that men living in Antarctica and during journey period exposed to both cold and emotional stress and allowed a free choice of diet would show some increase in there serum cholesterol. A comparatively lower level of total cholesterol at 2nd week stage in trial drug subjects could be due to presence of food supplement. The level of triglyceride showed a different behaviour. The higher level of triglycerides in trial drug group can be due to inclusion of highly nutritious oily food supplement in the diet. The levels decreased at later stages, which is in accordance with the studies by Satija et al (2000). Vallerand et al (1988)

also observed a decrease in triglycerides levels in men exposed to cold. The level of LDL was either lower or nearly equal at all stages in treatment group subjects could be due to some beneficiary effect of trial drug, which included antioxidants like Vitamin E and Vitamin C. The level of VLDL was almost equal in almost all the subjects at all stages of sampling. At 2nd week stage it slightly increased in treatment group & then observed a decrease at later stages. Radomski and Orme (1971) have observed that intramuscular triglyceride utilization and VLDL turnover are both increased during shivering in rats. A similar observation was made by Thenault and Poe (1965). Beneficiary effect of trial drug on HDL, HDL/LDL ratio may be due to its components. Cinnamomum has been shown to lower blood levels of fat and bad cholesterol. Green tea leaves has also been shown to lower cholesterol however the mechanism is not still known. Almonds contain 90% PUFA and 0% cholesterol. It has LDL lowering and HDL increasing property. The beneficial effects can be seen as early as two weeks. According to another study whole almond consumption reduced LDL, triglycerides and total cholesterol level whereas it increased HDL levels (Chopra, 2004). *Zingiber officinalis* has been shown to be hypo-lipidemic and anti-atherosclerotic effect in cholesterol fed rabbits (Furhman, 2000). In this study the levels of antioxidant enzymes and antioxidants was higher in treatment group along with generation of free radical at a lower pace. So it is quite possible that these antioxidants must have helped in maintaining the levels of HDL/LDL high.

From the results it may be concluded that food supplement helped to keep up the antioxidant status of the body, provided better adaptation competence and despite of rich oily components exerted a positive effect on the lipid profile without any significant effect on kidney function, liver function, body weight, body mass index and lean body mass (Communicated data).

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